

Geography Ch-7 (Must-Do) — Arid / Desert Landforms (Wind Action)

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain
Aeolian erosion & deposition: deflation, abrasion; yardang, zeugen, inselberg, barchan, loess
CA anchor: Greening of the Thar Desert (irrigation-driven change)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — Wind Erosion & Depositional Desert Landforms (GC Leong / Majid)

GS-I • Physical Geography →
Aeolian (Desert)
Landforms
(Source: GC Leong, Majid Husain)

■ NEWS TRIGGER

Desert (aeolian) geomorphology underpins India's arid Thar — where wind erosion, dunes and desertification meet irrigation-driven greening.

■ INTRO & ORIGIN

In deserts, wind is the chief agent. It erodes by **deflation** (lifting/rolling loose material) and **abrasion** (sand-blasting rock), then deposits sand as **dunes** and fine dust as **loess**.

Sparse vegetation + dry, loose regolith let wind pick up grains. Coarse sand does the abrading near the ground; fine dust travels far and settles as loess beyond desert margins.

■ KEY DATA

Process	Landform	What it is
Deflation	Deflation hollow	Depression scooped by wind (e.g. Qattara, Sahara)
Abrasion	Zeugen / Yardang	Mushroom/tabular rock / streamlined ridge parallel to wind
Abrasion	Ventifact / Dreikanter	Wind-faceted pebble (three faces = dreikanter)
Weathering+wind	Inselberg / Mesa	Residual isolated hill / flat-topped upland
Deposition	Barchan / Seif	Crescentic dune / longitudinal dune
Deposition	Loess	Wind-blown fine dust plain beyond desert margins

■ STATIC THEORY

- **Deflation** lowers the land into **deflation hollows** by blowing away loose sand/dust; a lag of pebbles left behind forms a **desert pavement**.
- **Abrasion** (wind-driven sand) sculpts **zeugen** (mushroom/rock-mushroom) and **yardangs** (streamlined ridges with long axis parallel to the wind, inverted-ship shape).
- **Ventifacts** are wind-faceted pebbles; a three-faced one is a **dreikanter**. Residual hills are **inselbergs**; flat-topped uplands are **mesas/buttes**.
- **Depositional dunes**: crescent-shaped **barchans** (horns point downwind) and long ridge-like **seifs (longitudinal dunes)** aligned with the wind.
- **Loess** = extensive deposits of wind-blown fine silt/dust carried far beyond the desert (very fertile).

■ MUST-KNOW FACTS

1. Wind erosion = deflation + abrasion.
2. Deflation hollow + desert pavement (lag of pebbles).

■ UPSC TRAPS

- ✗ Barchan horns point into (upwind of) the wind.
- ✓ Barchan horns point DOWNWIND (leeward); the gentle slope faces the wind.

3. Yardang: long axis PARALLEL to wind; zeugen = rock mushroom. 4. Dreikanter = 3-faced wind-faceted pebble (ventifact). 5. Barchan = crescentic dune; seif = longitudinal dune. 6. Loess = wind-blown fertile silt beyond desert margins.	✗ A yardang's long axis is perpendicular to the wind. ✓ A yardang is streamlined PARALLEL to the wind direction. ✗ Loess is a water-laid deposit. ✓ Loess is WIND-blown fine dust (aeolian), not fluvial.
■ MAINS ANGLE <i>Explain deflation vs abrasion and classify desert landforms, applying them to India's Thar (dunes, wind erosion) and its dynamic greening/desertification.</i>	■ STUDY LINK Geography → Physical → Arid/Aeolian Landforms (GC Leong: Wind & Deserts; Majid: Aeolian Geomorphology)

■ MEDIUM UPSC RELEVANCE

MEDIUM CA ANCHOR — Greening of the Thar Desert	GS-I / GS-III · Aeolian geography applied → Land-use change
■ NEWS TRIGGER Remote-sensing studies show parts of the Thar 'greening' over two decades — driven by canal irrigation (Indira Gandhi Canal), farm expansion and rising rainfall — even as fringes still degrade.	
■ INTRO & ORIGIN The Thar shows that a 'desert' is dynamic: aeolian dune landscapes can be partly stabilised/greened by irrigation and land-use change, while other margins still face wind erosion.	
■ STATIC THEORY - Drivers of greening: Indira Gandhi Canal irrigation, ~300% farm expansion and 50-800% urbanisation in some areas (1985-2020). - Greening stabilises dunes locally, but the arid fringe (E/SE Thar) remains vulnerable to degradation. - Shows the difference between the physical 'sand desert' and productivity-based 'desertification'.	
■ MUST-KNOW FACTS 1. Thar greening driven mainly by Indira Gandhi Canal irrigation. 2. It is the world's most densely populated desert region. 3. Desert extent is dynamic — greening vs fringe degradation.	■ UPSC TRAPS ✗ Deserts are static and always expanding. ✓ Extent is dynamic — irrigation can green/stabilise dunes while other areas degrade.
■ MAINS ANGLE <i>Use aeolian geomorphology to discuss irrigation-driven Thar greening vs desertification trade-offs.</i>	■ STUDY LINK Geography → Arid Landforms → Desert Dynamics (applied CA)